

Corporate Bond Valuation Under Credit Risk

FICC Derivatives Trading Case Study

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Problem Setup

A **5-year unsecured corporate bond** pays a 6% annual coupon (Face Value $F = \$100$). The risk-free rate is 4% (flat). The conditional default probability is $\lambda = 5\%$ per year.

Key Formulas:

$$S(t) = (1 - \lambda)^t = 0.95^t \quad (\text{Survival Prob.}) \quad D(t) = (1 + r)^{-t} = 1.04^{-t} \quad (\text{Discount Factor})$$

Year	$S(t)$	$D(t)$	$S(t) \cdot D(t)$	$P_d(t) = S(t-1) \cdot \lambda$
1	0.9500	0.9615	0.9135	0.0500
2	0.9025	0.9246	0.8344	0.0475
3	0.8574	0.8890	0.7622	0.0451
4	0.8145	0.8548	0.6962	0.0429
5	0.7738	0.8219	0.6358	0.0407
Σ			3.8421	0.2262

Part (a): Bond Value with Zero Recovery

$$V = \sum_{t=1}^5 C \cdot S(t) \cdot D(t) + F \cdot S(5) \cdot D(5) = 6 \times 3.8421 + 100 \times 0.6358$$

Answer (a): $V = 23.05 + 63.58 = \$86.63$

Part (b): Credit Spread with Zero Recovery

Solve for YTM y where $86.63 = \sum_{t=1}^5 \frac{6}{(1+y)^t} + \frac{100}{(1+y)^5}$

y	9.0%	9.5%	9.48%	10.0%
Price	\$88.33	\$86.56	\$86.63	\$84.83

Answer (b): $s = y - r = 9.48\% - 4.00\% = \mathbf{548 \text{ bps}}$

Part (c): Bond Value with \$70 Recovery

With recovery $R = \$70$, add expected recovery: $PV_{rec} = \sum_{t=1}^5 R \cdot P_d(t) \cdot D(t)$

Year	$P_d(t)$	$R \times P_d(t) \times D(t)$
1	0.0500	$70 \times 0.0500 \times 0.9615 = 3.37$
2	0.0475	$70 \times 0.0475 \times 0.9246 = 3.07$
3	0.0451	$70 \times 0.0451 \times 0.8890 = 2.81$
4	0.0429	$70 \times 0.0429 \times 0.8548 = 2.57$
5	0.0407	$70 \times 0.0407 \times 0.8219 = 2.34$
Total		\$14.16

Answer (c): $V = 23.05 + 63.58 + 14.16 = \text{\$100.79}$

Part (d): Credit Spread with \$70 Recovery

Solve for YTM: $100.79 = \sum_{t=1}^5 \frac{6}{(1+y)^t} + \frac{100}{(1+y)^5}$

y	5.5%	5.8%	5.81%	6.0%
Price	\$102.16	\$100.84	\$100.79	\$100.00

Answer (d): $s = 5.81\% - 4.00\% = \text{181 bps}$

Table with all outcomes

Metric	Zero Recovery ($R = \$0$)	With Recovery ($R = \$70$)	Δ Change
Bond Value	\$86.63	\$100.79	+\$14.16 (+16.4%)
YTM	9.48%	5.81%	-3.67%
Credit Spread	548 bps	181 bps	-367 bps

Summary

1. Both scenarios share identical 5% default probability, yet spreads differ by 367 bps. This demonstrates that credit spreads primarily compensate for *loss severity* (LGD), not just default frequency.
2. With 70% recovery, the bond trades above par (\$100.79), indicating the 6% coupon adequately compensates for residual credit risk.
3. Accurate recovery modeling—through seniority analysis, collateral valuation, and covenant review—is essential for relative value trading and risk management in credit markets.

For your reference, the full source code for this analysis is available in the [GitHub Repository](#).